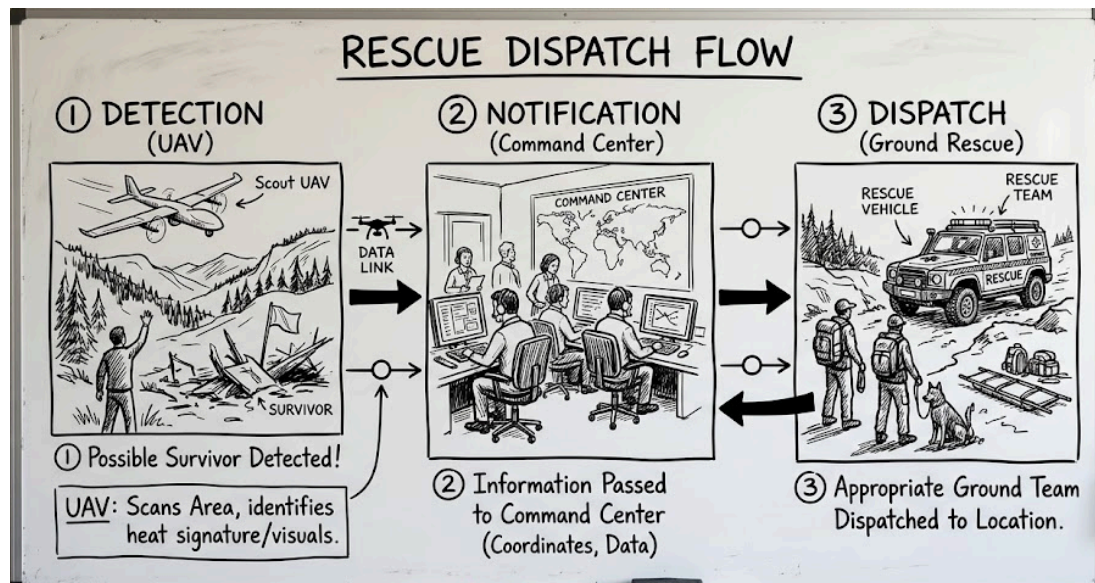




M7.4 earthquake rocks Colombia, killing at least 111 people; Aug 10

Earthquake Response Using UAVs

In an ideal workflow, when an earthquake occurs there is a command center that is placed at a safe distance from the earthquake occurrence.



Command Center

Medical Team

- Medical Diagnose and Take to the Nearest Hospital

Rescue Team

- **Ground Rescue Team**
- Concrete Cutter, Jackhammer, Hydraulic Spreader, Air Lifting Bag, Crane. Rope System

Technical Search Team

- Search Camera (ফাঁক দিয়ে ভেতর দেখা)
- Thermal Camera
- Sensitive Microphone
- Life Detector
- Rescue Dogs (K9)
- **Search Drones *****

Command Center Basically Controls these teams and gives instructions .



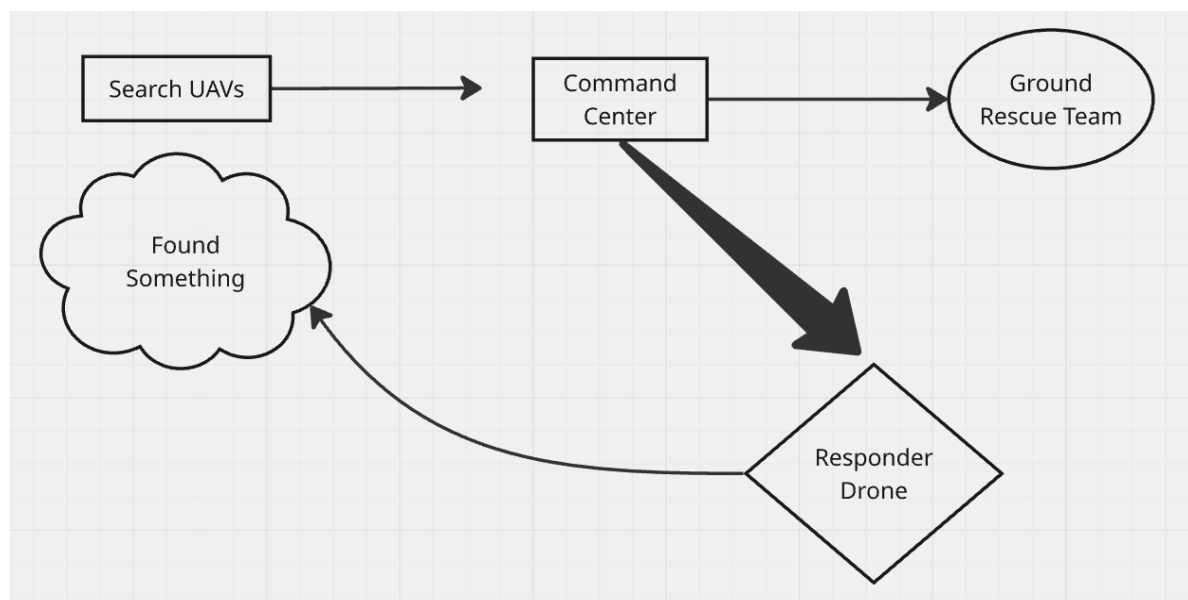
So, basically when the search drones find a new sign of survivor they inform it to the command center. Now the command center will allocate a ground rescue team to get those trapped humans out.

C#1 : Available	C#2 : Ground Response Team Not Available
They arrive at the location given by the search scan.	Wait until a ground rescue team gets free. Possible reasons include: (1) Collapsed buildings and rubble can make physical access difficult. (2) Roads may be blocked or unsafe. (3) A suitable rescue team may already be occupied with another incident. (4) Multiple survivor locations may require attention at the same time.

Usually this problem is so severe that it takes hours, sometimes days to reach a location that has been scanned as a sense of life.

This is a research gap and here we can **use another UAV to respond**.

The responder UAV is not intended to replace the human rescue team. Instead, it acts as a “golden-hour bridge” between detection and human response.



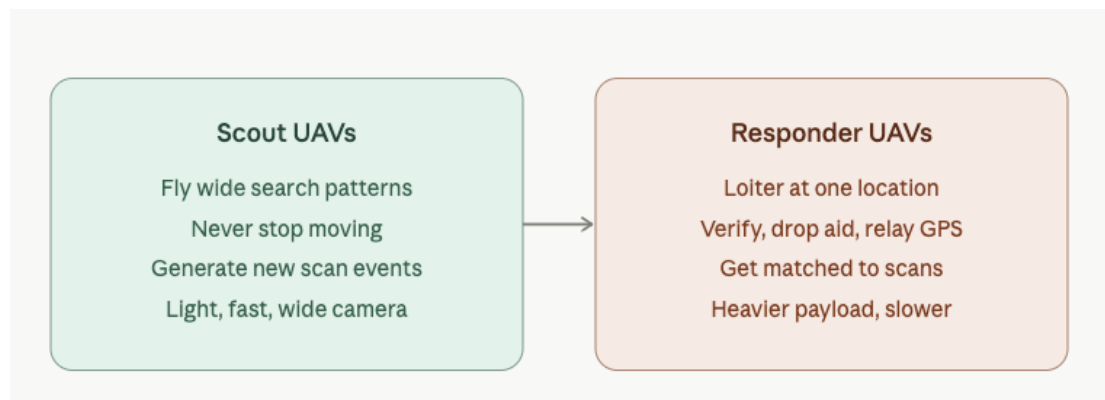
What is the Job of the Rescue Drone ? (Service Model)

Opt1 : Drops Medical Supply / First Aid/ Medicine / Oxygen/Statlite Phn/Radio

Opt2 : Hovers Over the Survivor and

- Streams Live Video
- Tries to talk with microphone with Ground Rescue Team(Drones propeller makes huge sound , so this might be a problem)
- Updates Exact Location

Will the rescue UAV keep hovering near the survivor, or will it return immediately after delivering the medicine so it can also serve other victims?



A few concrete reasons the two roles genuinely can't be the same drone:

- **Different hardware, different job.** A scout drone is built **light** and **fast**, with a wide-angle camera, optimized for endurance and area coverage. A responder drone might need to carry a medkit or supplies, hover steadily for a closer look, or relay a live feed/GPS ping back to a ground rescue team — that's heavier, slower, different equipment entirely. Trying to make one drone do both usually makes it worse at each.
- **Stopping breaks the search.** A scout drone flying a coverage pattern is optimized to keep moving and sweep as much ground as possible. If it stops every time it spots someone — to loiter, verify, and assist — it stops scanning the rest of its assigned area. Every stop means it's *not* finding the next survivor. You'd trade "more thorough response" for "worse overall coverage," which is a bad trade in a scenario where many people need to be found quickly. It's also not going to be fair, because that would basically give unfair advantage to the person that is being found first.

The Core Problem

When a scan drone sends some location to the command center. Now , we may have multiple responder drones available. Which drone to send to which location. This becomes an **Offloading Problem**.

The Responder Drones also need a path to go to the location it has to the shortest. It may be currently servicing a particular location or could be at the command center itself. Now the command center needs to calculate the distance that is able to be filled in the energy. This becomes a **Routing Problem**.

Offloading Problem Constraints	Routing problem Constraints
- Battery Limited - Can be in the Command Center or could be serving another	- Battery

But the scenario makes the offloading and the routing problem non trivial.

Need an Online Algorithm (Offloading)

This current scenario is an Online problem. Because,

1. **A new scan arrives** — a drone just spotted someone. The system asks: "Given everyone's current position/battery, who should go?"
2. **A drone becomes free** — it finished (or aborted) its last job. The system asks: "is there an unmatched, still-relevant scan this drone should pick up now?"

So it's not "solve once, done" — it's a rolling, event-triggered recalculation, running at the ground station/edge server (Portable Server) , and it needs to be fast (seconds, not minutes) since drones are already burning battery and survivors are already waiting while it computes. The travel time (minutes) is the slow part; the matching decision itself should be comparatively instant.

This calculation is done in the command center.

Also if there are multiple survivors who gets the first

Time-Dependent Urgency (Offloading)

A person trapped under rubble for 5 minutes and one trapped for 3 hours are not equally urgent — the value of reaching them drops the longer they wait (that's the "golden hour" idea). So each scan isn't just a fixed task, it's a task whose importance is quietly ticking down while it waits to be matched. In the diagram, Scan C fading in urgency, and Scan B still waiting, versus Scan A, which is fresh and high-value.

Uncertainty : The detection itself might be wrong (Routing)

When a drone's camera flags "possible survivor," that's not a guaranteed fact — it's a guess with some confidence attached. It could be a false alarm (rubble shaped like a person), or the drone could miss a real survivor entirely. The idea: instead of treating every scan as a certain, equal task, your system should **trust high-confidence detections more than low-confidence ones** — maybe sending a full responder drone immediately for a highly-confident scan, but sending a quick second look (re-verification) for a shaky one before committing a scarce responder to it. This matters a lot in your specific scenario because false positives are a real, documented problem in thermal-imagery survivor detection — it's not a hypothetical uncertainty, it's one people actually run into

Possible Extension :

Energy-bounded recourse: normally online matching irrevocable (একবার match করলে change করা যায় না)। কিন্তু বাস্তবে যদি একটা high-urgency survivor হঠাৎ পাশে দেখা যায়, drone-কে reroute করা reasonable। "Online matching with recourse" নিজেই একটা hot theoretical area, কিন্তু তোমার domain-specific twist: recourse free না — প্রতিটা reassignment battery খরচ করে, আর battery একটা shared, finite, non-renewable-mid-mission resource। এটা "bounded recourse" কে "cost-bounded recourse" বানায় — genuinely fresh combination।

- > Scout Drones and Responder Drones must not crash into each other.
- > Energy Problem : (15-30 mins of flight) If the Responder UAV keeps flying back and forth to drop kits, it will run out of juice fast. Design a "Hot-Swap Battery Station" on a nearby fire truck, so the drone lands, a human swaps the battery in 10 seconds, and it takes off again.
- > Relay UAVs : Communication bridge in GPS-denied areas Mesh networking capability
- > How would the drones travel in a 3D map where the height also matters.
- > During an earthquake, aftershocks happen dynamically while the drone is en route. A path that was clear 2 minutes ago might be completely buried by a secondary collapse. (Dynamic Re-Routing on-the-fly)
- > "Fairness Threshold" in your urgency model. For example: "No survivor with a severity score above 0.8 will wait longer than 10 minutes without a Responder"